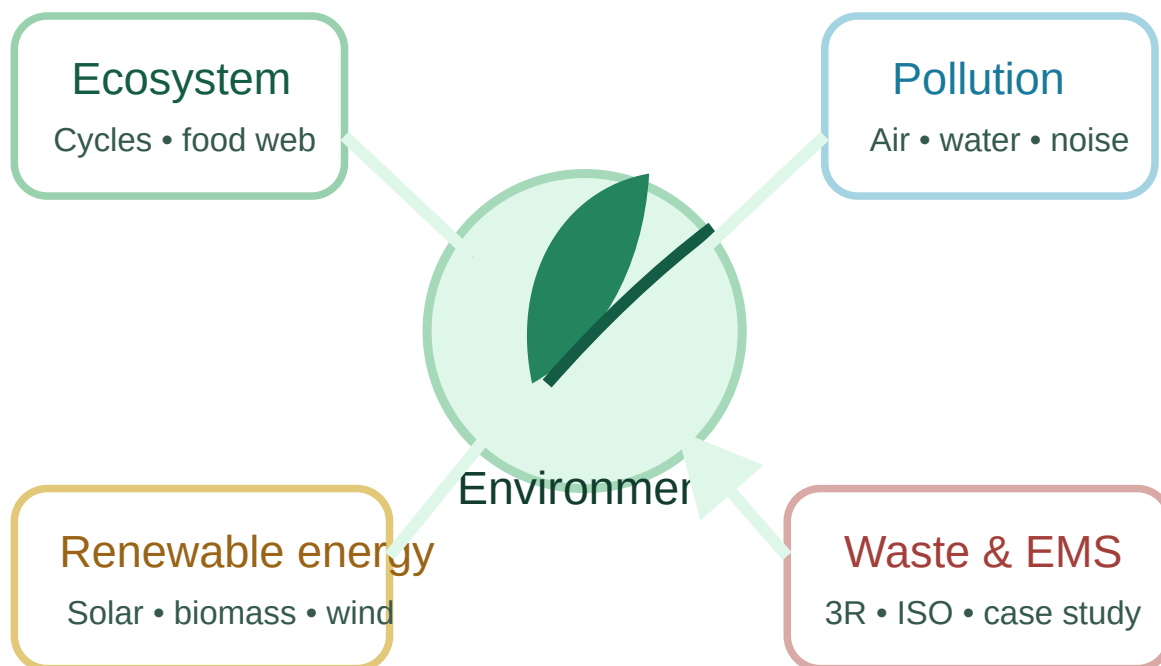


# Environmental Science

A complete handbook for ecosystems, pollution control, renewable energy, solid-waste management, environmental law awareness, ISO 14000, carbon accounting and field-based case studies—with Malayalam support, worked problems, diagrams, revision tools and answer keys.

**Course Code****2001****Semester****2****Credits****No Credit****Category****Common Course****Periods/Week****3 (L:2 T:1 P:0)****Total Periods****45****Course Outcomes****4****Series Tests****2 × 1 hour**

**Understand systems. Measure impacts. Design better solutions.**



Module 1 → Module 2 → Module 3 → Module 4 → Case study and sustainable engineering action

#### COURSE PURPOSE

## Objectives and prerequisite

### Understand ecosystems

Develop essential knowledge of ecosystem structure, energy transfer, nutrient cycles and global environmental processes.

### Recognise environmental problems

Study the sources, effects and control of air, water, soil and noise pollution.

### Apply technical thinking

Use engineering observation, measurement and design skills to propose workable environmental solutions.

### Explore renewable energy

Understand solar, biomass, wind, hydrogen, ocean, tidal and geothermal resources and their harvesting processes.

# Manage materials and waste

Study municipal, industrial, electronic, biomedical and hazardous waste, including 3R, recovery and disposal.

## Prerequisite

Basic Chemistry from Semester 1 Applied Chemistry supports understanding of gases, water quality, reactions, fuels and materials.

**പഠനരീതി:** ഓരോ environmental topic-വും “source → pathway → effect → measurement → control → prevention” എന്ന ക്രമത്തിൽ പഠിച്ചാൽ ഓർക്കാൻ എളുപ്പമാണ്.

### OFFICIAL OUTCOME STRUCTURE

## Course outcomes and duration

| CO   | Outcome                                                                                                                                                         | Hours | Cognitive level |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|
| CO 1 | Explain the ecosystem and terminology involved in it.                                                                                                           | 10    | Understanding   |
| CO 2 | Explain air, water, soil and noise pollution, and control measures and acts.                                                                                    | 11    | Understanding   |
| CO 3 | Explain different renewable energy resources and efficient processes of harvesting.                                                                             | 9     | Understanding   |
| CO 4 | Explain solid-waste management, ISO 14000 and environmental management, and conduct a case study on an environmental problem or sustainable-energy application. | 13    | Understanding   |

**Hour verification:** 10 + 11 + 9 + 13 = 43 teaching hours. Two one-hour series tests complete the 45-period semester.

### CO-PO MAPPING

## Outcome relationship

| Course outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1            | —   | —   | —   | —   | —   | —   | —   |
| CO2            | —   | —   | —   | —   | 2   | —   | —   |
| CO3            | —   | —   | —   | —   | 3   | —   | —   |
| CO4            | —   | —   | —   | —   | —   | 3   | 3   |

In the syllabus legend, 3 means strongly mapped, 2 moderately mapped and 1 weakly mapped.

### DETAILED SYLLABUS MAP

# Module-by-module handbook structure

Course 2002

| Module code    | Official topic                                                   | Hours | Handbook expansion                                                                  |
|----------------|------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------|
| M1.01          | Structure of ecosystem; biotic and abiotic components            | 1     | Levels of organisation, habitats, ecological pyramids, examples and system diagrams |
| M1.02          | Food chain and food web                                          | 1     | Trophic levels, energy transfer, grazing/detritus chains and stability              |
| M1.03          | Aquatic lentic/lotic and terrestrial ecosystem                   | 1     | Pond, lake, river, forest, grassland and desert comparison                          |
| M1.04          | Carbon, nitrogen, sulphur and phosphorus cycles                  | 4     | Processes, reservoirs, human disturbances, diagrams and tracing problems            |
| M1.05          | Global warming, greenhouse effect, ozone depletion and acid rain | 3     | Mechanisms, effects, prevention and misconception correction                        |
| M2.01          | Pollution/pollutant; natural and man-made air-pollution sources  | 3     | Source–pathway–receptor model; refrigerants, IC engines and boilers                 |
| M2.02          | Air pollutants and control equipment                             | 3     | Particulates, gases, bag filter, cyclone, ESP, absorber and catalytic converter     |
| M2.03          | Water pollution and pH, DO, BOD, COD                             | 3     | Parameters, interpretation, sampling awareness and solved comparisons               |
| M2.04          | Noise pollution and rules                                        | 2     | Sources, decibel scale, effects, controls and logarithmic examples                  |
| Series Test I  | Modules 1 and 2                                                  | 1     | Original practice test with complete answers                                        |
| M3.01          | Solar energy and collectors                                      | 2     | Flat-plate collectors, coatings, pond, heater, dryer and still                      |
| M3.02          | Biomass and biogas                                               | 2     | Fuel properties, anaerobic digestion, storage and utilisation                       |
| M3.03          | Wind energy                                                      | 2     | Resource, turbine parts, Indian context, benefits and challenges                    |
| M3.04          | Hydrogen, ocean, tidal and geothermal energy                     | 3     | Principles, conversion routes, applications and comparison                          |
| M4.01          | Municipal, e-waste and biomedical waste                          | 2     | Sources, characteristics, risks and segregation                                     |
| M4.02          | Metallic and non-metallic industrial waste                       | 2     | Lubricants, plastics, rubber, metals and recovery options                           |
| M4.03          | Collection/disposal, 3R, recovery, landfill and hazardous waste  | 2     | Waste hierarchy, logistics, landfill anatomy and treatment selection                |
| M4.04          | Syllabus-listed environmental acts and pollution-control boards  | 1     | Purpose, institutional roles and current-source verification guidance               |
| M4.05          | Carbon credit and carbon footprint                               | 1     | Scopes, activity data, factors and solved footprint example                         |
| M4.06          | Environmental management in fabrication industry                 | 1     | Aspect–impact register, controls, monitoring and improvement                        |
| M4.07          | ISO 14000 implementation and benefits                            | 1     | EMS cycle, ISO 14001 concepts, documentation and audit                              |
| M4.08          | Case study                                                       | 3     | Problem definition, data, engineering solution, sustainability and report           |
| Series Test II | Modules 3 and 4                                                  | 1     | Original practice test with complete answers                                        |

## HOW TO USE THIS HANDBOOK

# Six-step learning workflow

1 Read the concept and identify all key terms.

2 Study the system diagram or process flow.

3 Follow one worked example or engineering scenario.

**4** Answer the quick-check questions without looking back.

**5** Attempt a case-based or numerical problem.

**6** Use the revision, viva and master-answer sections for final preparation.

**Answer-writing pattern:** Definition → source/process → effect → control/prevention → labelled diagram or example → short conclusion.

## Module 1 — Ecosystem and Environmental Processes

Understand ecosystem structure, energy transfer, aquatic and terrestrial systems, biogeochemical cycles and major atmospheric environmental problems.

**M1.01 • 1 HOUR**

### Structure of an ecosystem

#### Abiotic components

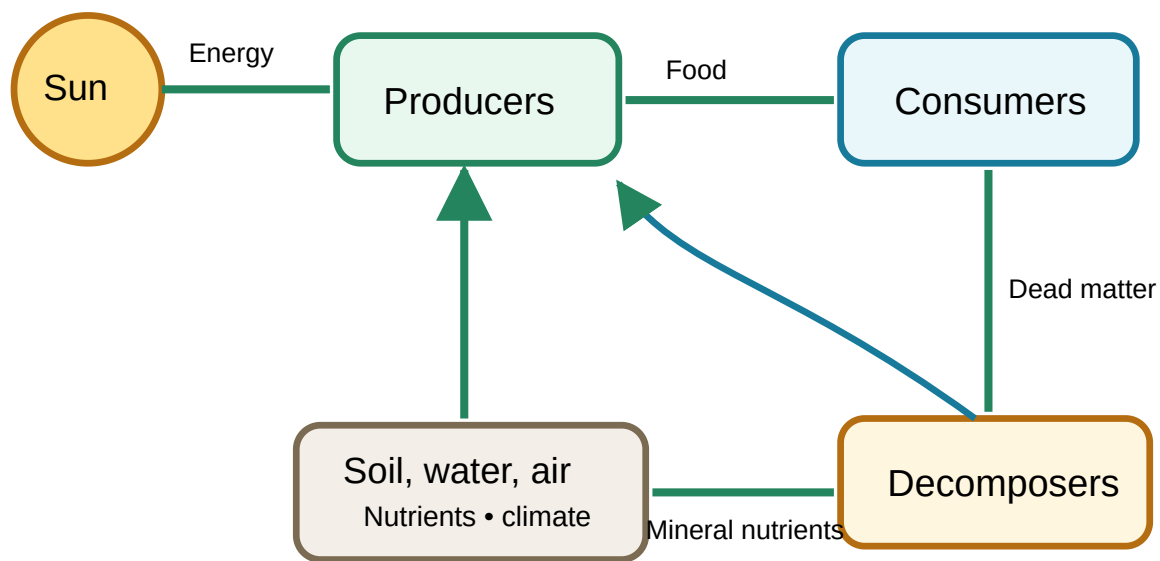
Non-living physical and chemical factors: sunlight, temperature, water, air, soil, minerals, pH, salinity and climate.

#### Biotic components

Living organisms grouped functionally as producers, consumers and decomposers.

#### System interaction

Energy enters mainly as sunlight, moves through organisms and leaves as heat, while matter cycles repeatedly.



Energy flows mainly in one direction; nutrients are recycled by decomposition and biogeochemical processes.

## Functional groups

| Group                        | Role                                                 | Examples                  | Engineering relevance                              |
|------------------------------|------------------------------------------------------|---------------------------|----------------------------------------------------|
| Producers                    | Convert solar or chemical energy into organic matter | Green plants, algae       | Biomass production, oxygen release, carbon capture |
| Primary consumers            | Feed directly on producers                           | Deer, rabbit, zooplankton | Transfer plant energy upward                       |
| Secondary/tertiary consumers | Feed on other consumers                              | Frog, snake, eagle        | Control populations and maintain balance           |
| Decomposers                  | Break down dead matter and wastes                    | Bacteria, fungi           | Nutrient recycling, composting and treatment       |

## Ecological organisation

### Organism

One individual living being.

### Population

Individuals of one species in an area.

### Community

Interacting populations.

### Ecosystem

Community plus abiotic environment.

### Biome

Large climatic ecological region.

**Biosphere**  
All ecosystems on Earth.

## Ecological pyramids

### Pyramid of numbers

Shows the number of organisms at each trophic level. It may be upright or inverted depending on the ecosystem.

### Pyramid of biomass

Shows living mass at each trophic level. It can be inverted in some aquatic systems because phytoplankton renew rapidly.

### Pyramid of energy

Shows energy transfer per unit area and time. It is always upright because energy is lost as heat at every transfer.

#### Worked Problem 1

### Energy available across trophic levels

A grassland producer level captures 20,000 kJ. Assuming a simplified 10% transfer at each level, estimate energy available to primary, secondary and tertiary consumers.

Primary =  $20,000 \times 0.10 = 2,000$  kJ

Secondary =  $2,000 \times 0.10 = 200$  kJ

Tertiary =  $200 \times 0.10 = 20$  kJ

**Interpretation:** Large producer energy supports much less energy at the top. This limits the number and biomass of top predators.

ഓരോ trophic level-ലും വലിയൊരു ഭാഗം energy heat ആയി നഷ്ടപ്പെടുന്നതിനാൽ energy pyramid എല്ലായ്പ്പോഴും upright ആണ്.

#### M1.02 • 1 HOUR

## Food chain and food web

### Food chain

A linear feeding sequence showing transfer of matter and energy from one organism to another.

Grass → Grasshopper → Frog → Snake → Eagle

# Food web

A network of interconnected food chains. It better represents real ecosystems because organisms usually have multiple food sources and predators.

Grass → Rabbit → Fox  
 Grass → Grasshopper → Frog → Snake  
 Seeds → Mouse → Snake → Eagle

## Types of food chain

| Type                | Starts with         | Example                        | Importance                           |
|---------------------|---------------------|--------------------------------|--------------------------------------|
| Grazing food chain  | Living green plant  | Grass → deer → tiger           | Transfers freshly fixed solar energy |
| Detritus food chain | Dead organic matter | Leaf litter → earthworm → bird | Recycles wastes and dead biomass     |

**Why food webs improve stability:** If one food source declines, some organisms may shift to alternatives. However, highly specialised species may still be vulnerable.

### Scenario Removing a predator

If snakes are heavily removed from a crop-field food web, rodent numbers can rise. More rodents may damage crops and consume stored grain. The problem is not only “one species missing”; it is a cascade through several links.

**Engineering lesson:** Pest management should consider ecosystem interactions rather than relying only on repeated chemical control.

M1.03 • 1 HOUR

## Aquatic and terrestrial ecosystems

| Ecosystem | Key condition                              | Typical producers                    | Typical consumers                  | Major environmental pressure                    |
|-----------|--------------------------------------------|--------------------------------------|------------------------------------|-------------------------------------------------|
| Lentic    | Standing water: pond, lake, reservoir      | Phytoplankton, algae, aquatic plants | Zooplankton, fish, insects         | Eutrophication, sedimentation, oxygen depletion |
| Lotic     | Flowing water: stream, river               | Attached algae, riparian vegetation  | Insects, fish, amphibians          | Flow alteration, sewage, sand mining            |
| Forest    | Tree-dominated terrestrial system          | Trees and understory plants          | Insects, birds, mammals            | Deforestation and fragmentation                 |
| Grassland | Grass-dominated with periodic grazing/fire | Grasses and herbs                    | Grazers and predators              | Overgrazing and conversion                      |
| Desert    | Very low rainfall and high stress          | Xerophytes and seasonal plants       | Reptiles, insects, adapted mammals | Water extraction and habitat disturbance        |



**Lentic versus lotic:** Lentic systems often develop vertical zones and may stratify. Lotic systems are shaped by flow velocity, channel form, oxygen exchange and upstream activities.

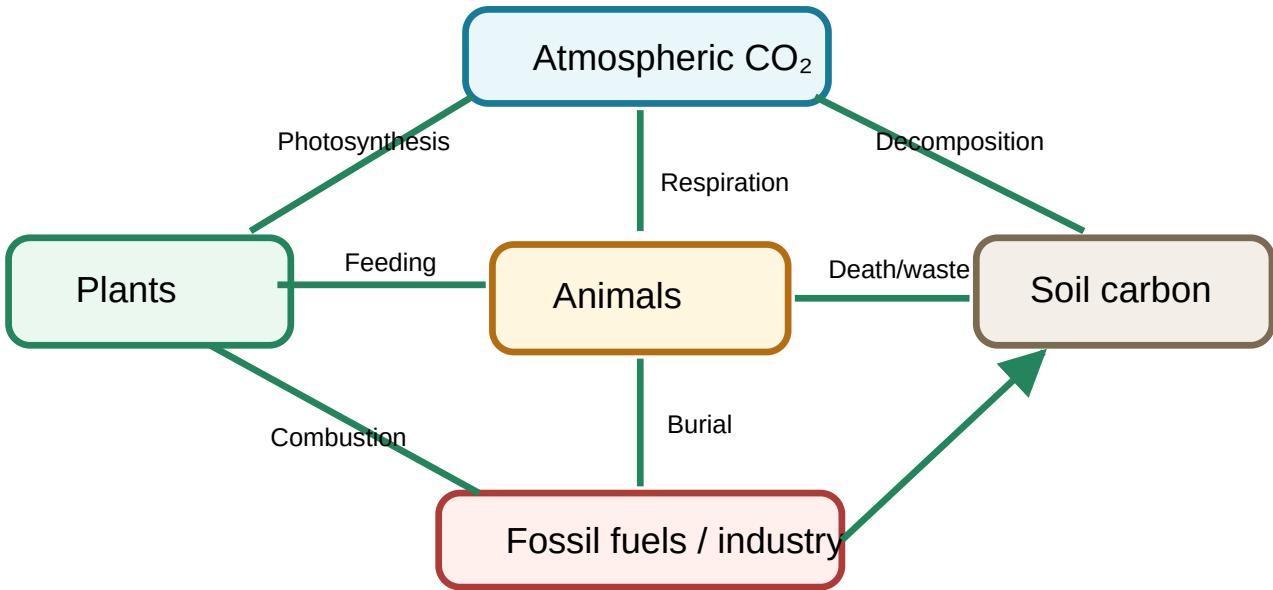
**ശാർക്ക:** Lentic = standing water; Lotic = flowing water. Pond/lake lentic ആണ്, stream/river lotic ആണ്.

M1.04 • 4 HOURS

## Biogeochemical cycles

Biogeochemical cycles move essential elements among atmosphere, water, soil, rocks and organisms. Human activity can change the speed, location and form of these transfers.

### Carbon cycle



Photosynthesis removes atmospheric CO<sub>2</sub>; respiration, decomposition and combustion return it.

| Process        | Direction                     | Human influence                             |
|----------------|-------------------------------|---------------------------------------------|
| Photosynthesis | Atmosphere → biomass          | Deforestation reduces uptake                |
| Respiration    | Biomass → atmosphere          | Natural biological process                  |
| Decomposition  | Dead matter → soil/atmosphere | Landfill conditions may produce methane     |
| Combustion     | Fuel/biomass → atmosphere     | Fossil-fuel burning increases rapid release |
| Ocean exchange | Atmosphere ↔ ocean            | Warming and acidification alter balance     |

### Nitrogen cycle

#### Nitrogen fixation

Converts atmospheric N<sub>2</sub> into biologically usable forms through microbes, lightning or industrial processes.

## Assimilation

Plants absorb nitrate/ammonium and build proteins; animals obtain nitrogen by feeding.

## Ammonification

Decomposers convert organic nitrogen in wastes and dead matter into ammonia/ammonium.

## Denitrification

Microbes convert nitrate back to gaseous forms under low-oxygen conditions.

## Human disturbance

Excess fertiliser and sewage can cause nutrient enrichment, algal blooms and nitrate contamination.

# Sulphur cycle

Sulphur occurs in rocks, soil, water, living matter and atmospheric gases. Weathering releases sulphates; plants assimilate them; food transfers sulphur; decomposition returns it. Volcanic emissions and combustion of sulphur-containing fuels release  $\text{SO}_2$ , which can form acids in the atmosphere.

# Phosphorus cycle

Phosphorus has no major gaseous phase under normal conditions. Weathering releases phosphate from rocks, plants absorb it, animals obtain it by feeding, and decomposition returns it to soil and water. Runoff carrying fertiliser or detergent phosphate can intensify eutrophication.

| Cycle      | Main reservoir                    | Important processes                                    | Typical disturbance                               |
|------------|-----------------------------------|--------------------------------------------------------|---------------------------------------------------|
| Carbon     | Atmosphere, ocean, rocks, biomass | Photosynthesis, respiration, combustion                | Fossil-fuel use and deforestation                 |
| Nitrogen   | Atmosphere                        | Fixation, nitrification, assimilation, denitrification | Excess fertiliser and reactive nitrogen emissions |
| Sulphur    | Rocks, sediments and ocean        | Weathering, assimilation, decomposition, oxidation     | Fuel combustion and mining                        |
| Phosphorus | Rocks and sediments               | Weathering, uptake, food transfer, sedimentation       | Runoff and eutrophication                         |

### Worked Problem 2 Tracing fertiliser nitrogen

A field receives 100 kg of nitrogen. Suppose 55 kg is taken up by crops, 15 kg remains in soil, 10 kg is lost to the atmosphere and the rest leaves through runoff/leaching.

$$\text{Runoff/leaching} = 100 - (55 + 15 + 10) = 20 \text{ kg}$$

**Interpretation:** The 20 kg loss can affect groundwater or surface water. Better timing, dose control, soil testing and buffer strips can reduce loss.

M1.05 • 3 HOURS

# Global warming, greenhouse effect, ozone depletion and acid rain

## Natural greenhouse effect and enhanced warming

The atmosphere allows much incoming shortwave solar radiation to reach Earth. The warmed surface emits longwave infrared radiation. Greenhouse gases absorb and re-emit part of this outgoing energy, keeping the planet warmer than it would otherwise be. Human activities raise concentrations of several greenhouse gases, strengthening this effect.

| Gas               | Major human-related sources                                | Environmental relevance                                 |
|-------------------|------------------------------------------------------------|---------------------------------------------------------|
| Carbon dioxide    | Fossil-fuel combustion, cement production, land-use change | Major long-lived driver of warming                      |
| Methane           | Oil/gas systems, livestock, rice, landfill, wetlands       | Strong warming influence and ozone precursor            |
| Nitrous oxide     | Fertiliser, manure, industry and combustion                | Long-lived greenhouse gas and ozone-depleting influence |
| Fluorinated gases | Refrigeration, electronics and industry                    | Some have very high warming potential                   |

## Causes

- Combustion of coal, oil and gas
- Deforestation and land-use change
- Industrial processes and cement
- Agriculture and livestock
- Waste decomposition and refrigerant leakage

## Effects

- Rising average temperature and heat extremes
- Changes in rainfall and drought/flood risk
- Ice loss and sea-level rise
- Ecosystem and crop stress
- Health, infrastructure and livelihood impacts

**Common misconception:** The greenhouse effect is not the same as ozone depletion. They involve different gases, mechanisms and consequences, although some substances can influence both.

## Ozone depletion

Stratospheric ozone absorbs harmful ultraviolet radiation. Certain long-lived chlorine- and bromine-containing compounds can reach the stratosphere, where radiation releases reactive atoms that catalytically destroy ozone. Consequences include increased UV exposure, skin and eye damage, and effects on crops and plankton.

## Acid rain

Sulphur dioxide and nitrogen oxides can react in the atmosphere to form sulphuric and nitric acids. Wet deposition occurs in rain, fog or snow; dry deposition occurs as gases and particles. Effects include acidification, nutrient loss from soil, aquatic stress and corrosion of structures.

| Problem                 | Main mechanism                                               | Control direction                                                                  |
|-------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|
| Enhanced global warming | Higher greenhouse-gas concentration traps more outgoing heat | Efficiency, renewable energy, low-carbon transport, methane control, carbon sinks  |
| Ozone depletion         | Catalytic destruction of stratospheric ozone                 | Phase-out controlled substances, recover refrigerants, prevent leakage             |
| Acid rain               | SO <sub>2</sub> and NO <sub>x</sub> form acids in atmosphere | Cleaner fuels, flue-gas control, low-NO <sub>x</sub> combustion, transport control |

### Worked Problem 3 Simple emissions comparison

Option A consumes 1,000 kWh from a source with an illustrative emission factor of 0.70 kg CO<sub>2</sub>e/kWh. Option B uses 650 kWh at the same factor.

$$A = 1,000 \times 0.70 = 700 \text{ kg CO}_2\text{e}$$

$$B = 650 \times 0.70 = 455 \text{ kg CO}_2\text{e}$$

$$\text{Illustrative reduction} = 700 - 455 = 245 \text{ kg CO}_2\text{e}$$

**Important:** Emission factors vary by place, time and method. Use the current authorised factor for real reporting.

## Module 1 quick check

### 1. Why is energy flow unidirectional while nutrients cycle?

Energy is progressively dissipated as heat, whereas atoms are conserved and reused through biological, geological and chemical processes.

### 2. Differentiate a food chain and food web.

A food chain is one linear feeding path; a food web is a network of interconnected feeding paths.

### 3. Why can excess phosphorus cause eutrophication?

It can remove a nutrient limitation, encouraging algal growth. Decomposition of excess biomass consumes oxygen.

### 4. Greenhouse effect versus ozone depletion?

The greenhouse effect concerns heat retention by infrared-absorbing gases; ozone depletion concerns loss of stratospheric UV protection.

## Module 2 — Pollution, Measurement and Control

**M2.01 • 3 HOURS**

## Pollution, pollutants and sources

### Pollution

An undesirable change in the physical, chemical or biological quality of the environment that harms living organisms, materials, comfort or useful environmental functions.

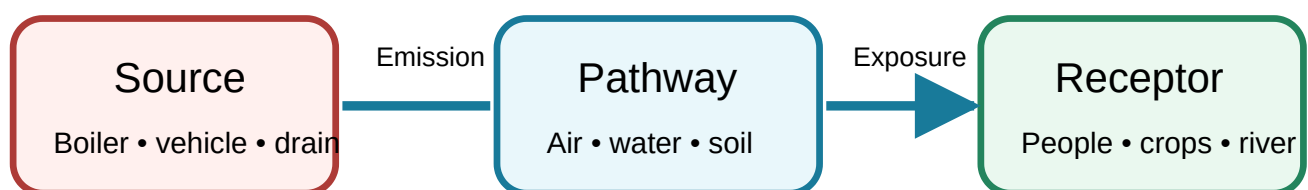
### Pollutant

A substance, form of energy or biological agent present at a concentration, duration or location capable of causing adverse effects.

### Receptor

The person, organism, ecosystem, material or process that receives exposure and may be affected.

## Source–pathway–receptor model



Control can act at the source, interrupt the pathway or protect the receptor; source prevention is usually preferred.

## Natural and human-made sources

| Source<br>Course 2001<br>class | Examples                                                    | Typical pollutants                                             | Control approach                                                                 |
|--------------------------------|-------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------|
| Natural                        | Dust storms, wildfire, sea spray, volcanic activity, pollen | Particles, gases, aerosols, biological matter                  | Monitoring, warning, exposure reduction and land management                      |
| Transport                      | Internal-combustion vehicles                                | CO, NO <sub>x</sub> , hydrocarbons, particles, CO <sub>2</sub> | Maintenance, cleaner fuels, catalytic control, public transport, electrification |
| Boilers/furnaces               | Industrial and commercial combustion                        | Particles, SO <sub>2</sub> , NO <sub>x</sub> , CO and ash      | Fuel quality, combustion optimisation and flue-gas treatment                     |
| Refrigeration                  | Leaks during operation or servicing                         | Refrigerants with ozone or climate impacts                     | Leak prevention, recovery, correct charging and approved alternatives            |
| Construction                   | Excavation, crushing and transport                          | Dust and diesel exhaust                                        | Water suppression, covers, housekeeping and equipment maintenance                |
| Agriculture                    | Fertiliser, livestock, residue burning                      | Ammonia, methane, nitrous oxide, smoke                         | Nutrient planning, manure management and no-burn alternatives                    |

## Primary and secondary pollutants

**Primary pollutants** are emitted directly, such as CO, SO<sub>2</sub> or dust. **Secondary pollutants** form in the environment through reactions, such as ground-level ozone and some secondary particles.

**മുഖ്യകക:** Source-ൽ നേരിട്ട് പുറപ്പെടുന്നത് primary pollutant; atmosphere-ൽ reaction കഴിഞ്ഞ് രൂപപ്പെടുന്നത് secondary pollutant.

M2.02 • 3 HOURS

# Air pollutants, effects and control equipment

## Particulate and gaseous pollutants

| Pollutant             | Typical source                           | Effect                                                    | Suitable control direction                               |
|-----------------------|------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| Coarse/fine particles | Combustion, grinding, road dust          | Respiratory/cardiovascular stress, visibility and soiling | Cyclone, bag filter, ESP, enclosure and suppression      |
| SO <sub>2</sub>       | Sulphur-containing fuel                  | Respiratory irritation, acid deposition and corrosion     | Low-sulphur fuel, desulphurisation and absorption        |
| NO <sub>x</sub>       | High-temperature combustion              | Ozone/smog formation and respiratory impacts              | Combustion control, catalytic treatment                  |
| CO                    | Incomplete combustion                    | Reduces oxygen transport in blood                         | Complete combustion, ventilation and catalytic oxidation |
| VOCs/hydrocarbons     | Solvents, fuel and incomplete combustion | Odour, toxicity and ozone formation                       | Substitution, capture, adsorption, oxidation             |
| Ground-level ozone    | Secondary photochemical formation        | Lung and crop damage                                      | Control NO <sub>x</sub> and VOC precursors               |

## Particle-control devices

Control Device 1

### Cyclone separator

Dust-laden gas enters tangentially and forms a vortex. Centrifugal action drives heavier particles toward the wall; particles lose momentum and fall into a hopper while cleaner gas exits.

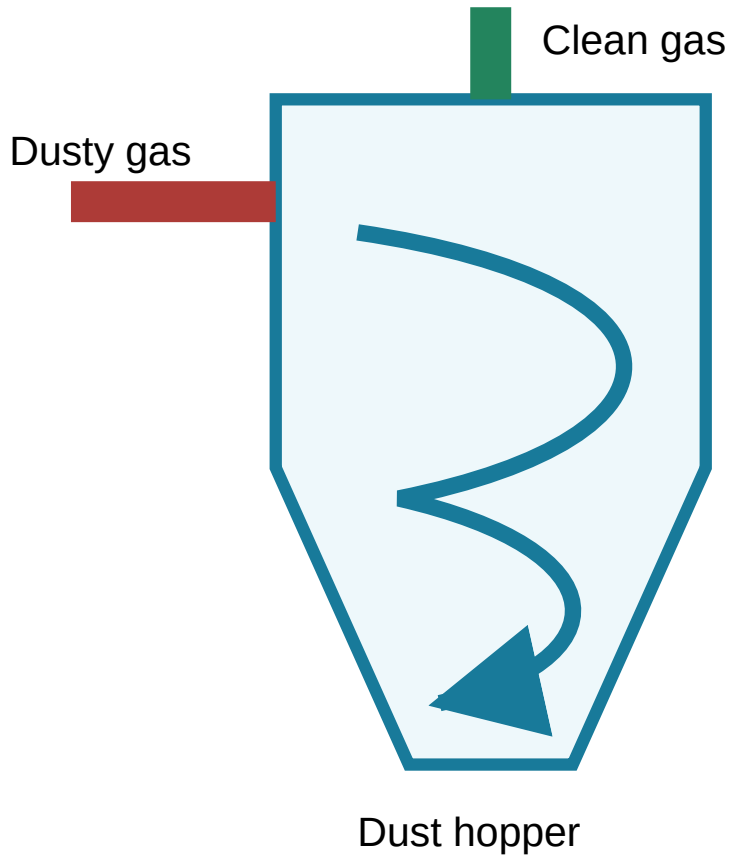
**Strengths:** simple, robust, low pressure loss, suitable as pre-cleaner.

**Limitations:** less efficient for very fine particles.

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**Control Device 2****Bag filter**

Gas passes through fabric bags. Particles are captured on fibres and in the dust cake. Periodic shaking, reverse air or pulse cleaning removes dust into a hopper.

**Best suited**

High removal of many dry particles, including fine dust.

**Watch for**

Temperature, moisture, corrosiveness and fire/explosion risk.

**Maintenance**

Bag integrity, differential pressure and dust disposal.

**Control Device 3****Electrostatic precipitator (ESP)**

Particles are electrically charged and attracted to oppositely charged collection plates. Rapping dislodges collected material into hoppers.

**Strengths:** high efficiency for large gas flows with relatively low pressure drop. **Limitations:** performance depends on particle electrical properties, installation and operating conditions.

## Gaseous-pollution control

### Absorber / scrubber

Transfers a gaseous pollutant from gas into a liquid. Selection depends on solubility and reaction chemistry. A packed tower provides large contact area.

**Example:** alkaline liquid absorbing an acidic gas.

### Catalytic converter

Uses catalysts to promote oxidation and reduction reactions in vehicle exhaust, reducing CO, unburned hydrocarbons and NO<sub>x</sub> when the engine and fuel-control system operate correctly.

**Important:** It does not eliminate CO<sub>2</sub> produced from fuel combustion.

## Control-device selection

| Situation                                        | Likely first choice | Reason                               |
|--------------------------------------------------|---------------------|--------------------------------------|
| Large, coarse, dry particles before another unit | Cyclone             | Robust pre-cleaning                  |
| Fine dry dust with suitable temperature          | Bag filter          | High filtration efficiency           |
| Very large flue-gas flow                         | ESP                 | Large capacity and low pressure drop |
| Soluble/reactive gas                             | Absorber            | Gas-to-liquid mass transfer          |
| Vehicle CO/HC/NO <sub>x</sub>                    | Catalytic converter | Catalysed exhaust reactions          |

### Worked Problem 4 Collection efficiency and outlet concentration

A dust-control unit receives gas containing 500 mg/m<sup>3</sup> of particulate matter and removes 92%.

$$\text{Removed concentration} = 500 \times 0.92 = 460 \text{ mg/m}^3$$

$$\text{Outlet concentration} = 500 - 460 = 40 \text{ mg/m}^3$$

**Check:** 40/500 = 0.08, so 8% remains and 92% is removed.

**Air-quality standards:** Standards set concentration limits or assessment methods for specified pollutants and averaging periods. Real compliance work must use the current official standard, approved sampling method and calibrated instruments.



# Water pollution and quality parameters

Course 2002

## Sources and effects

| Source              | Possible contaminants                              | Effect                                   | Prevention/control                               |
|---------------------|----------------------------------------------------|------------------------------------------|--------------------------------------------------|
| Domestic sewage     | Organic matter, nutrients, pathogens               | Oxygen depletion and disease risk        | Collection, treatment and sanitation             |
| Industry            | Acids/alkalis, metals, solvents, heat and organics | Toxicity, pH change and ecosystem damage | Cleaner production and effluent treatment        |
| Agricultural runoff | Nutrients, sediment and pesticides                 | Eutrophication and contamination         | Correct application, buffers and erosion control |
| Urban runoff        | Oil, litter, metals and sediment                   | Short pollution pulses and flooding      | Drain maintenance and stormwater management      |
| Solid-waste dumping | Leachate and plastics                              | Ground/surface-water pollution           | Segregation and engineered disposal              |

## Important parameters

### pH

A logarithmic measure related to hydrogen-ion activity. pH below 7 is acidic, 7 neutral under standard classroom simplification, and above 7 alkaline.

### Dissolved oxygen (DO)

Oxygen dissolved in water and available to aquatic organisms. Low DO may indicate high oxygen demand, poor mixing or warm/stagnant conditions.

### Biochemical oxygen demand (BOD)

Oxygen consumed by microorganisms while decomposing biodegradable organic matter under specified test conditions.

### Chemical oxygen demand (COD)

Oxygen-equivalent measure of substances chemically oxidised by a strong reagent under the test method.

### Turbidity

Cloudiness caused by suspended and colloidal particles, affecting light penetration and treatment.

### Nutrients and solids

Nitrogen, phosphorus, dissolved solids and suspended solids influence eutrophication, salinity and treatment needs.

| Pattern<br>Course 2001       | Likely interpretation                                                  |
|------------------------------|------------------------------------------------------------------------|
| High BOD + low DO            | Strong biodegradable organic load may be consuming oxygen              |
| COD much higher than BOD     | Substantial chemically oxidisable material may be poorly biodegradable |
| High nutrients + algal bloom | Eutrophication risk                                                    |
| Extreme pH                   | Possible industrial/chemical input and toxicity                        |
| High turbidity               | Sediment, runoff, disturbance or treatment problem                     |

Worked Problem 5

## Interpreting two water samples

| Sample | DO (mg/L) | BOD (mg/L) | COD (mg/L) |
|--------|-----------|------------|------------|
| A      | 7.2       | 2          | 12         |
| B      | 2.1       | 28         | 75         |

**Interpretation:** Sample B shows much higher oxygen demand and much lower dissolved oxygen, indicating a substantially greater pollution load than A. COD exceeds BOD in both because the chemical method can measure a wider range of oxidisable substances.

**Do not diagnose from one number alone:** Sampling location, temperature, test method, time and other parameters must be considered.

Worked Problem 6

## Simple BOD removal efficiency

An influent has BOD 180 mg/L and treated effluent has BOD 27 mg/L.

Removal efficiency =  $(180 - 27) / 180 \times 100 = 85\%$

The treatment removed 153 mg/L BOD-equivalent load from the measured concentration basis.

## Soil pollution awareness

Although the detailed outline emphasises air, water and noise, the course outcome also names soil pollution. Soil can be contaminated by uncontrolled dumping, oil, heavy metals, pesticides, salts and industrial residues. Effects include reduced fertility, food-chain transfer, groundwater contamination and direct exposure.

**Parameter memory:** DO കൂടുതലാകുന്നത് പൊതുവെ നല്ല oxygen availability സൂചിപ്പിക്കും; BOD/COD ഉയരുന്നത് pollution load കൂടുതലാകാമെന്ന് സൂചിപ്പിക്കും.

M2.04 • 2 HOURS

## Noise pollution, measurement and control

Noise is unwanted or harmful sound. Its impact depends on sound level, frequency, duration, timing, impulsiveness, distance and receiver sensitivity.

## Effects

Hearing damage, sleep disturbance, stress, reduced concentration, communication interference and wildlife disturbance.

## Control hierarchy

Quiet source/equipment → enclosure or barrier → distance/layout → maintenance → scheduling → hearing protection.

## Decibel scale

Sound-pressure level is logarithmic. Therefore, two equal sound sources do not add arithmetically in decibels.

$$L_t = 10 \log_{10}(10^{(L_1/10)} + 10^{(L_2/10)} + \dots)$$

### Worked Problem 7 Combining two equal noise sources

Two independent machines each produce 80 dB at a point.

$$L_t = 10 \log_{10}(10^8 + 10^8) = 10 \log_{10}(2 \times 10^8)$$

$$L_t \approx 83 \text{ dB}$$

Two equal sources increase the level by about 3 dB, not to 160 dB.

### Worked Problem 8 Effect of distance in a simplified free field

For a point source in an ideal free field, doubling distance reduces sound level by approximately 6 dB. If a machine produces 92 dB at 2 m, an approximate level at 4 m is 86 dB and at 8 m is 80 dB. Real rooms, reflections and multiple sources alter this estimate.

## Control at source, path and receiver

| Location | Examples                                                                      |
|----------|-------------------------------------------------------------------------------|
| Source   | Quieter equipment, balancing, lubrication, reduced speed, vibration isolation |
| Path     | Enclosures, acoustic barriers, absorptive lining, increased distance          |
| Receiver | Cabins, work rotation, limiting exposure, suitable hearing protection         |

**Regulatory awareness:** The syllabus lists the Noise Pollution (Regulation and Control) Rules, 2000. For actual compliance, use the current official text, amendments, zone classification, measurement protocol and competent-authority guidance.

# Original practice test — Modules 1 and 2

This is a syllabus-based practice test, not an official board question paper.

1. Define ecosystem and distinguish biotic and abiotic components.
2. Explain food chain, food web and ecological pyramid.
3. Draw and explain any one biogeochemical cycle.
4. Differentiate greenhouse effect, ozone depletion and acid rain.
5. Define pollution and pollutant using the source–pathway–receptor model.
6. Explain working, advantage and limitation of a cyclone or bag filter.
7. Differentiate DO, BOD and COD.
8. A treatment plant reduces BOD from 240 mg/L to 36 mg/L. Calculate percentage removal.
9. Explain effects and control of noise pollution.
10. Two equal sources produce 85 dB each. Estimate the combined level.

## Module 3 — Renewable Energy Resources and Harvesting

Solar, biomass, wind, hydrogen, ocean, tidal and geothermal energy: principles, equipment, applications, calculations, benefits and limitations.

### M3.01 • 2 HOURS

## Solar energy

Solar systems capture radiation, convert it into heat or electricity, transfer or store it and deliver it to a load. Output depends on location, season, time, cloud, orientation, tilt, shading and losses.

### Direct radiation

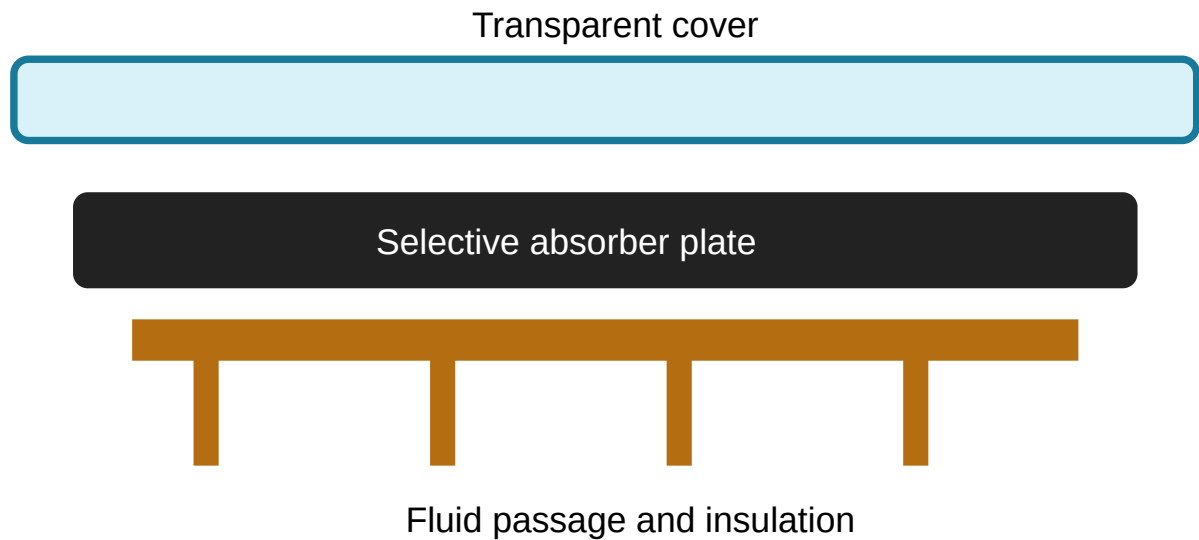
Reaches a surface without atmospheric scattering; concentrating collectors mainly require it.

### Diffuse radiation

Scattered by air, cloud and particles; flat-plate collectors can receive it.

### Global radiation

Total direct and diffuse radiation received by the surface.



Cover reduces convective loss; the dark absorber converts radiation to heat; tubes or air channels carry heat; insulation reduces rear loss.

| System             | Principle                                               | Application                               | Limitation                          |
|--------------------|---------------------------------------------------------|-------------------------------------------|-------------------------------------|
| Liquid flat plate  | Water/antifreeze flows through tubes on absorber        | Water and low-temperature process heating | Scaling, freezing and leakage       |
| Air flat plate     | Air passes over/under heated absorber                   | Space and crop drying                     | Lower heat capacity and duct losses |
| Solar water heater | Collector transfers heat to stored water                | Domestic/institutional hot water          | Cloud and backup requirement        |
| Solar dryer        | Solar-heated air removes moisture                       | Food/crop preservation                    | Temperature-humidity control        |
| Solar still        | Evaporation followed by condensation                    | Small-scale distilled water               | Low daily output                    |
| Solar pond         | Salinity gradient suppresses convection and stores heat | Low-grade heat                            | Area and gradient maintenance       |
| Advanced collector | Concentrates direct radiation                           | Higher-temperature heat                   | Tracking and clear-sky dependence   |

### Worked Problem 9 Solar water-heating energy

Raise 100 L of water from 25°C to 60°C. Take density 1 kg/L and specific heat 4.18 kJ/kg·°C.

$$Q = m c \Delta T = 100 \times 4.18 \times 35 = 14,630 \text{ kJ} = 14.63 \text{ MJ}$$

At 50% overall efficiency, simplified incident energy required =  $14.63/0.50 = 29.26 \text{ MJ}$ .

### Worked Problem 10 Collector output

Area 4 m<sup>2</sup>, irradiance 700 W/m<sup>2</sup> and efficiency 45%.

$$\text{Useful power} = 4 \times 700 \times 0.45 = 1,260 \text{ W} = 1.26 \text{ kW}$$

**Solar memory:** Coating radiation absorb ചെയ്യുകയും thermal radiation loss കുറയ്ക്കുകയും ചെയ്യുന്നു.

### M3.02 • 2 HOURS

## Biomass and biogas

Biomass is renewable organic material from plants, animals and biodegradable residues. Moisture, ash, volatile matter, fixed carbon, calorific value and bulk density affect fuel behaviour.

### Anaerobic digestion stages

- 1 Hydrolysis:** complex material becomes soluble molecules.
- 2 Acidogenesis:** products become organic acids, alcohols,  $H_2$  and  $CO_2$ .
- 3 Acetogenesis:** intermediates become acetate,  $H_2$  and  $CO_2$ .
- 4 Methanogenesis:** microorganisms produce methane-rich biogas.

### Biogas use

Cooking, heating, engine-generator fuel after cleaning, or upgrading to biomethane.

### Digestate

May return nutrients when pathogens and contaminants are properly managed.

### Key controls

Temperature, pH, loading, retention time, mixing, toxicity, gas tightness and safety.

### Worked Problem 11 Estimated gas yield

Feed = 20 kg/day; illustrative yield = 0.04 m<sup>3</sup>/kg.

$$\text{Biogas} = 20 \times 0.04 = 0.80 \text{ m}^3/\text{day}$$

At 60% methane, methane volume = 0.48 m<sup>3</sup>/day. Actual yield depends on substrate and operating condition.

**Safety:** Biogas is flammable and may contain toxic/corrosive H<sub>2</sub>S. Provide ventilation, leak control, pressure relief and trained operation.

M3.03 • 2 HOURS

## Wind energy

A wind turbine converts air-stream kinetic energy into rotor motion and electricity. Main parts are blades, hub, nacelle/drivetrain, generator, controls, tower and foundation.

$$\text{Available wind power} = \frac{1}{2} \rho A v^3$$

Actual output also includes the power coefficient and mechanical/electrical efficiencies. Because power varies with the cube of wind speed, site measurement is essential.

Worked Problem 12

### Approximate turbine output

$\rho = 1.225 \text{ kg/m}^3$ ,  $A = 50 \text{ m}^2$ ,  $v = 8 \text{ m/s}$  and power coefficient = 0.35.

$$P = \frac{1}{2} \times 1.225 \times 50 \times 8^3 \times 0.35 \approx 5,488 \text{ W} = 5.49 \text{ kW}$$

| Benefits                            | Challenges and controls                         |
|-------------------------------------|-------------------------------------------------|
| No fuel combustion during operation | Variable output needs grid/storage planning     |
| Low operational water use           | Visual, acoustic and shadow issues need siting  |
| Land may retain agricultural use    | Wildlife risk needs assessment and mitigation   |
| Scalable technology                 | Roads, foundations and maintenance have impacts |

**ശ്രദ്ധിക്കുക:** Wind speed ചെറിയ തോതിൽ മാറിയാലും power വലിയ തോതിൽ മാറാം, കാരണം  $v^3$  ബന്ധമാണ്.

M3.04 • 3 HOURS

## Hydrogen, ocean, tidal and geothermal energy

| Source        | Origin/conversion                                                                  | Application                     | Main challenge                                             |
|---------------|------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------|
| Hydrogen      | Produced by electrolysis, reforming or other routes; used in fuel cells/combustion | Industry, transport and storage | Production emissions, efficiency, cost, storage and safety |
| Wave          | Wind-generated wave motion drives a device                                         | Coastal electricity             | Storm survival and maintenance                             |
| Tidal range   | Sea-level difference drives barrage/lagoon turbines                                | Predictable electricity         | Ecology and sediment change                                |
| Tidal stream  | Underwater current drives turbine                                                  | Predictable marine power        | Installation and marine-use conflict                       |
| Ocean thermal | Surface/deep-water temperature difference drives heat engine                       | Power and possible co-products  | Low efficiency and large seawater flow                     |
| Geothermal    | Earth heat supplies steam/hot water                                                | Power and direct heating        | Location, drilling, scaling and induced effects            |

## Dry steam

Natural steam drives a turbine directly.

## Flash steam

Hot pressurised water is depressurised so part flashes to steam.

## Binary cycle

Geothermal fluid heats a separate low-boiling working fluid.

### Worked Problem 13

## Electrolyser electricity

An illustrative electrolyser uses 52 kWh per kg hydrogen. For 100 kg:

$$\text{Electricity} = 52 \times 100 = 5,200 \text{ kWh}$$

Life-cycle emissions depend strongly on the electricity source.

**Selection principle:** Match resource to load, location, scale, skill, cost, reliability, maintenance and environmental safeguards. A hybrid solution may be better than one technology.

## Module 3 quick check

### Why is selective coating important?

It combines strong solar absorption with reduced thermal radiation loss.

### Name four anaerobic-digestion stages.

Hydrolysis, acidogenesis, acetogenesis and methanogenesis.

### Why measure wind at the site?

Wind varies with location, height, terrain and time, and power is highly sensitive to speed.

### Is hydrogen a primary source?

It is generally an energy carrier; energy is required to produce it.



# Module 4 — Solid Waste, Environmental Management and ISO 14000

Explain waste sources and characteristics, segregation, collection, recovery, disposal, carbon concepts, institutional roles, fabrication-industry management and environmental management systems.

## M4.01 • 2 HOURS

## Municipal solid waste, e-waste and biomedical waste

Waste management begins with correct identification and segregation. Mixing incompatible or hazardous material with ordinary waste increases exposure, treatment difficulty and cost.

| Waste stream          | Typical source                               | Characteristics                                                  | Main risk                                                                  | Preferred management direction                                                         |
|-----------------------|----------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Municipal solid waste | Homes, shops, institutions and streets       | Food waste, paper, plastics, glass, metals, inert matter         | Odour, vectors, leachate, litter and methane                               | Source segregation, collection, composting/recycling, recovery and engineered disposal |
| E-waste               | Discarded electrical/electronic equipment    | Metals, plastics, glass, circuit boards, batteries               | Lead, mercury and other hazardous constituents; unsafe informal processing | Authorised collection, repair/reuse, producer systems and controlled recycling         |
| Biomedical waste      | Healthcare diagnosis, treatment and research | Infectious waste, sharps, contaminated material, pharmaceuticals | Infection, injury and chemical exposure                                    | Segregation at source, labelled containers, trained handling and authorised treatment  |

## Important waste characteristics

### Physical

Density, moisture, particle size, composition and compactability affect collection and processing.

### Chemical

Calorific value, ash, carbon, nutrients, metals, pH and reactivity affect recovery and treatment.

### Biological

Biodegradability, pathogens and putrescibility influence composting, odour and health risk.

**Segregation principle:** Sharps, batteries, chemicals, infectious waste and ordinary recyclables must not be mixed. Real handling must follow current local and sector-specific rules.

## Daily municipal-waste estimate

A campus has 800 users and an observed generation rate of 0.35 kg/person·day.

$$\text{Daily waste} = 800 \times 0.35 = 280 \text{ kg/day}$$

If 45% is biodegradable, biodegradable fraction =  $280 \times 0.45 = 126 \text{ kg/day}$ . This estimate helps size segregation, collection and treatment systems.

**ശ്രദ്ധിക്കുക:** Waste treatment-ന് മുമ്പ് source segregation നിർബന്ധം. Mixed waste ആയാൽ recycling/composting quality കുറഞ്ഞുപോകും.

### M4.02 • 2 HOURS

## Metallic and non-metallic industrial wastes

| Waste                         | Fabrication/industrial source       | Concern                                                   | Management option                                                        |
|-------------------------------|-------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|
| Metal offcuts and chips       | Cutting, machining and punching     | Sharp edges, oil contamination and loss of valuable metal | Segregate by alloy, drain fluids, reuse/recycle                          |
| Grinding dust                 | Grinding and finishing              | Fine particles and possible metal exposure                | Local extraction, sealed collection and authorised recovery/disposal     |
| Spent lubricants              | Machines, gearboxes and maintenance | Soil/water contamination and fire risk                    | Prevent leaks, labelled storage and authorised recovery                  |
| Plastic scrap                 | Packaging, moulding and components  | Persistence, contamination and mixed polymer streams      | Reduction, clean segregation and suitable recycling                      |
| Rubber waste                  | Seals, belts, tyres and offcuts     | Fire, storage and slow degradation                        | Retreading/reuse, material recovery or authorised treatment              |
| Paint/sludge/solvent residues | Surface preparation and coating     | Flammability, VOCs and hazardous constituents             | Substitution, process control, closed containers and authorised disposal |

## Material-flow thinking

For each input, track quantity purchased, quantity in product, reusable return, recyclable scrap, emissions and disposal. A material balance can reveal avoidable loss.

## Metal utilisation

A workshop purchases 1,000 kg sheet metal. Product contains 780 kg, clean segregated offcuts are 170 kg, and mixed/unaccounted loss is 50 kg.

$$\text{Product yield} = 780 / 1,000 \times 100 = 78\%$$

$$\text{Recoverable offcut rate} = 170 / 1,000 \times 100 = 17\%$$

Improvement may include nesting optimisation, better segregation, chip recovery, purchase-size selection and accurate weighing.

**M4.03 • 2 HOURS**

# Collection, 3R, energy recovery, landfill and hazardous waste

## 3R principles

### Reduce

Avoid generating waste through design, purchasing, process control and efficient use.

### Reuse

Use an item or component again with little processing, after safety and quality checks.

### Recycle

Process segregated material into secondary raw material or products.

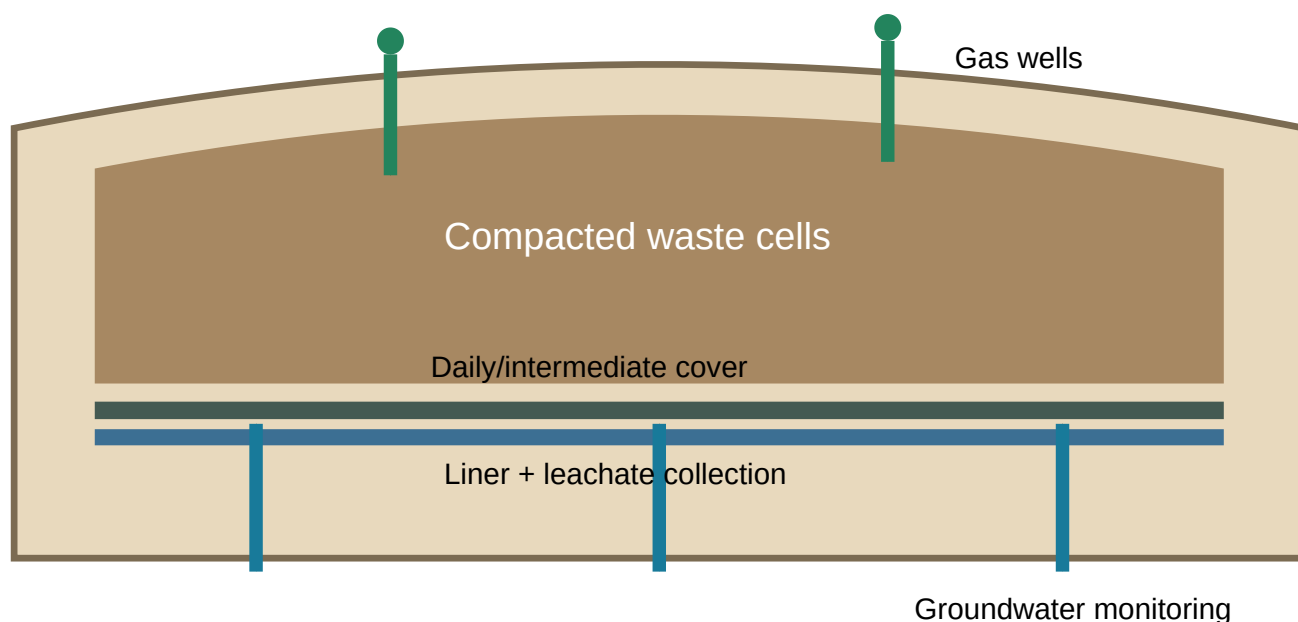
**Hierarchy:** Prevention/reduction is generally preferred over reuse, recycling, recovery and disposal because it avoids upstream resource and energy use.

## Collection system

- 1 Characterise waste and generation points.
- 2 Provide colour/labelled containers suited to the stream.
- 3 Set collection frequency based on quantity and risk.
- 4 Use safe internal transport and prevent spillage.
- 5 Record weight, destination and responsible person.
- 6 Send each stream only to an authorised next step.

## Energy recovery

## Sanitary landfill



An engineered landfill uses liners, leachate collection, compacted cells, cover, gas management, drainage and monitoring.

## Hazardous-waste principles

Hazardous waste may be toxic, flammable, corrosive, reactive, infectious or otherwise dangerous. It requires identification, compatible packaging, labels, secondary containment, manifests/records, trained handling and authorised treatment/disposal.

### Worked Problem 16 Landfill-volume estimate

Residual waste is 12 tonnes/day. Compacted density is 600 kg/m<sup>3</sup>. Estimate daily waste volume before cover.

$$\text{Mass} = 12,000 \text{ kg/day}$$

$$\text{Volume} = \text{mass/density} = 12,000/600 = 20 \text{ m}^3/\text{day}$$

If cover and operational allowance add 15%, planned daily volume  $\approx 23 \text{ m}^3/\text{day}$ .

M4.04 • 1 HOUR

## Environmental legislation awareness and pollution-control boards

The syllabus lists air-quality and pollution-control acts/rules and asks students to understand the structure and role of central and state pollution-control boards. The exact titles/years shown in teaching documents may contain legacy wording or typographical errors, so real compliance work must verify the current official legal text.

## Central-level functions

- National coordination and technical guidance
- Standards, monitoring programmes and data support
- Advising government and supporting laboratories
- Coordination with state boards and committees

## State-level functions

- Consent/authorisation functions under applicable law
- Inspection, monitoring and enforcement
- Review of pollution-control systems
- Complaint response, data collection and local implementation

**Legal-use caution:** This handbook is educational. For a project, consent application or compliance decision, use the current official CPCB/SPCB or government source and obtain competent professional guidance.

## What a student should understand

- 1 Environmental law defines responsibilities, standards, authorities and enforcement mechanisms.
- 2 Industries may require approvals, monitoring, records and reporting.
- 3 Boards inspect, sample, issue directions and guide pollution control within their authority.
- 4 Evidence must be based on approved methods, calibrated instruments and documented records.

M4.05 • 1 HOUR

## Carbon footprint and carbon credit

### Carbon footprint

A carbon footprint is the total greenhouse-gas emissions associated with an organisation, activity, product or person, expressed as carbon-dioxide equivalent (CO<sub>2</sub>e) over a defined boundary and period.

#### Direct emissions

Fuel burned in owned/controlled equipment.

#### Purchased-energy emissions

Electricity, steam, heat or cooling purchased.

#### Other indirect emissions

$$\text{Emissions} = \text{Activity data} \times \text{Applicable emission factor}$$

### Worked Problem 17 Illustrative monthly footprint

A workshop uses 200 kWh electricity at an illustrative factor of 0.70 kg CO<sub>2</sub>e/kWh and 20 L diesel at 2.31 kg CO<sub>2</sub>e/L.

$$\text{Electricity} = 200 \times 0.70 = 140 \text{ kg CO}_2\text{e}$$

$$\text{Diesel} = 20 \times 2.31 = 46.2 \text{ kg CO}_2\text{e}$$

$$\text{Total} = 186.2 \text{ kg CO}_2\text{e}$$

These factors are examples only. Real inventories need current authorised factors, clear boundaries and transparent assumptions.

## Carbon credit

A carbon credit generally represents a quantified greenhouse-gas reduction or removal under a defined programme. Credible credits depend on additionality, baseline, measurement, verification, permanence, leakage control and prevention of double counting.

**Reduction before offset:** First measure emissions, reduce avoidable energy/material use and improve technology. Credits do not replace direct reduction.

### M4.06 • 1 HOUR

## Environmental management in fabrication industry

### Typical aspects and impacts

| Activity          | Environmental aspect                       | Potential impact                    | Control                                                                |
|-------------------|--------------------------------------------|-------------------------------------|------------------------------------------------------------------------|
| Cutting/grinding  | Metal scrap, dust, noise and energy use    | Resource loss, air/noise exposure   | Optimised nesting, extraction, enclosure and maintenance               |
| Welding           | Fume, gases, electricity and electrodes    | Air exposure and resource use       | Process selection, local exhaust and housekeeping                      |
| Painting          | VOC, overspray, containers and sludge      | Air pollution and hazardous waste   | Low-VOC options, transfer efficiency, booth filters and closed storage |
| Machining         | Coolant, oil, chips and energy             | Water/soil pollution and metal loss | Leak prevention, fluid life extension and chip separation              |
| Cleaning          | Water, detergent and contaminated effluent | Water use and pollution load        | Counter-current cleaning, substitution and treatment                   |
| Material handling | Packaging and damaged material             | Waste and resource loss             | Reusable packaging and proper storage                                  |

### Aspect–impact evaluation

1 List activities, products and services.

2 Identify environmental aspects under normal, abnormal and emergency conditions.

3 Identify impacts and legal/other obligations.

4 Rank significance using a documented method.

5 Set operational controls, objectives and monitoring.

6 Review incidents, performance and improvement opportunities.

#### Worked Problem 18

### Simple coolant-loss improvement

A workshop purchased 500 L coolant concentrate per year. After leak repair and concentration control, use fell to 350 L with the same production.

$$\text{Reduction} = 500 - 350 = 150 \text{ L/year}$$

$$\text{Percentage reduction} = 150/500 \times 100 = 30\%$$

Benefits may include lower purchase cost, less waste, cleaner floors and reduced environmental risk.

#### M4.07 • 1 HOUR

## ISO 14000 and environmental management systems

ISO 14000 is a family of environmental-management standards. ISO 14001 specifies requirements for an environmental management system (EMS); it does not prescribe one universal emission level for every organisation.

### Plan–Do–Check–Act

#### Plan

Context, leadership, policy, aspects, obligations, risks, opportunities and objectives.

#### Do

Resources, competence, communication, documentation, operational control and emergency preparedness.

#### Check

## Act

Correct nonconformity and continually improve suitability, adequacy and effectiveness.

## Implementation sequence

1 Obtain leadership commitment and define EMS scope.

2 Develop environmental policy.

3 Identify aspects, impacts and compliance obligations.

4 Set measurable objectives and plans.

5 Establish competence, communication and documents.

6 Implement operational and emergency controls.

7 Monitor performance and evaluate compliance.

8 Conduct internal audits and management review.

9 Correct problems and improve continuously.

## Benefits

| Potential benefit              | How it arises                                               |
|--------------------------------|-------------------------------------------------------------|
| Better control                 | Clear responsibilities, procedures and records              |
| Lower resource use             | Objectives for energy, water and material efficiency        |
| Reduced risk                   | Aspect assessment and emergency preparedness                |
| Improved compliance management | Systematic tracking and evaluation of obligations           |
| Continual improvement          | Audit, review, corrective action and performance monitoring |

**ISO memory:** ISO 14001 certificate മാത്രം ലക്ഷ്യം അല്ല; day-to-day environmental control system പ്രവർത്തിക്കുകയാണ് പ്രധാനം.

M4.08 • 3 HOURS

## Case-study requirement

The syllabus states that the case study is compulsory for Continuous Internal Evaluation and not for the End Semester Examination. Students may work in groups of 4–5; topics should not be duplicated; each member submits the group report individually.



## Suggested direction 1

Engineering solution for a local environmental problem involving pollution, waste, water, noise, energy or material loss.

## Suggested direction 2

Study of a sustainable-energy resource with emphasis on a real application, suitability, benefits, limitations and improvement.

Full case-study workflow, sample reports, data tables and evaluation checklist are provided in the Case Study tab.

### SERIES TEST II • 1 HOUR

## Original practice test — Modules 3 and 4

This is not an official board question paper.

1. Explain the construction and working of a flat-plate collector.
2. Describe anaerobic digestion and biogas utilisation.
3. Calculate simplified wind power for given  $\rho$ ,  $A$ ,  $v$  and  $C_p$ .
4. Compare hydrogen, tidal and geothermal energy.
5. Explain sources and management of e-waste and biomedical waste.
6. Explain 3R and sanitary-landfill components.
7. Define carbon footprint and carbon credit.
8. Prepare an aspect–impact table for a fabrication workshop.
9. Explain PDCA in an ISO 14001 environmental management system.
10. Prepare a case-study outline for one local environmental problem.

## Environmental Case Study Handbook

A practical workflow for groups of 4–5 to investigate a real problem or sustainable-energy application and submit individual copies of the group report.

## Case-study workflow

**1 Select and approve the topic:** choose one local, observable problem or energy application that does not duplicate another group.

**2 Define the boundary:** location, activity, period, people/equipment and environmental media included.

**3 Write objectives:** specify what will be measured, compared or improved.

**4 Collect baseline evidence:** observation, count, weight, meter reading, interview, photograph, map or authorised record.

**5 Analyse source–pathway–receptor:** identify causes, transport routes and affected receptors.

**6 Generate alternatives:** prevention, reduction, reuse, control, treatment and monitoring options.

**7 Evaluate alternatives:** technical feasibility, environmental benefit, cost, safety, maintenance and acceptance.

**8 Recommend a solution:** include equipment/process, responsibilities, timeline and performance indicators.

**9 Prepare report:** use tables, diagrams, calculations, limitations, conclusion and references.

**10 Present and defend:** every group member must explain data, calculation and recommendation.

**Ethics and safety:** Do not enter restricted areas, touch hazardous waste, collect biological samples, publish personal data or operate equipment without permission and supervision.

## Required report structure

| Chapter                  | Required content                                                |
|--------------------------|-----------------------------------------------------------------|
| Cover page               | Course, title, institution, group, individual student and date  |
| Certificate/declaration  | Prescribed text and signatures where required                   |
| Abstract                 | Problem, method, major finding and recommendation in one page   |
| Introduction             | Background, significance and study location                     |
| Objectives and scope     | Clear questions, boundaries and limitations                     |
| Method                   | Observation, measurement, sampling or data-collection procedure |
| Results                  | Tables, figures, photographs and calculations                   |
| Discussion               | Interpretation, comparison and causes                           |
| Engineering solution     | Alternatives, selection criteria, design concept and controls   |
| Implementation plan      | Responsibilities, timeline, resources and indicators            |
| Conclusion               | Direct answer to objectives                                     |
| References and annexures | Sources, raw data, questionnaire and additional evidence        |

## Sample Case Study A — Campus biodegradable waste

### Problem statement

Mixed food waste from a campus canteen causes odour, attracts animals and contaminates recyclables. The study evaluates source segregation and a small composting/biogas option.

# Objectives

Course 2001

1. Estimate daily biodegradable waste.
2. Identify current handling gaps.
3. Compare composting and anaerobic digestion.
4. Recommend a safe, maintainable system.

## Seven-day demonstration dataset

| Day | Food preparation waste (kg) | Plate waste (kg) | Total (kg) |
|-----|-----------------------------|------------------|------------|
| 1   | 18                          | 12               | 30         |
| 2   | 17                          | 14               | 31         |
| 3   | 20                          | 13               | 33         |
| 4   | 19                          | 11               | 30         |
| 5   | 21                          | 15               | 36         |
| 6   | 16                          | 10               | 26         |
| 7   | 15                          | 9                | 24         |

Weekly total = 210 kg; average =  $210/7 = 30$  kg/day

If 80% is suitable for biological treatment after segregation, design basis = 24 kg/day. Add a safety margin only after observing seasonal and event variation.

## Alternatives

| Option                        | Strength                        | Limitation                                             |
|-------------------------------|---------------------------------|--------------------------------------------------------|
| Aerobic composting            | Simple, produces soil amendment | Requires bulking, aeration, moisture and odour control |
| Anaerobic digestion           | Produces biogas and digestate   | Higher process/safety skill and stable feed needed     |
| Authorised external treatment | Lower on-site operation         | Transport, service reliability and recurring cost      |

## Recommendation

Begin with two-bin segregation, weighing and contamination monitoring. Select composting or digestion only after space, operator skill, odour, water, digestate/compost use and approval requirements are verified.

## Performance indicators

- ✓ kg/day segregated
- ✓ contamination percentage
- ✓ odour/complaint count
- ✓ product quantity and quality
- ✓ downtime and maintenance
- ✓ residual waste sent for disposal

## Sample Case Study B — Solar hot water for a hostel

## Baseline

A hostel needs 1,000 L/day of water raised from 25°C to 55°C.

$$\text{Useful heat} = 1,000 \text{ kg} \times 4.18 \text{ kJ/kg} \cdot ^\circ\text{C} \times 30^\circ\text{C} = 125,400 \text{ kJ/day} = 125.4 \text{ MJ/day}$$

$$\text{Electrical equivalent} = 125.4/3.6 = 34.83 \text{ kWh/day}$$

If an existing electric heater system uses about 38 kWh/day including losses, annual energy for 300 operating days is 11,400 kWh.

## Study tasks

- 1 Measure actual hot-water quantity and temperature profile.
- 2 Check roof area, orientation, shading, structure and pipe route.
- 3 Estimate solar resource and system losses using approved design data.
- 4 Compare flat-plate and evacuated-tube systems where relevant.
- 5 Plan storage, backup, scaling control, insulation and maintenance.
- 6 Estimate energy and emissions avoided using current factors.
- 7 Assess life-cycle cost, safety and user behaviour.

**Conclusion format:** State technical suitability, expected useful-energy contribution, required collector/storage arrangement, controls, uncertainty and monitoring plan.

## Case-study submission checklist

### Evidence

- ✓ Dated observations
- ✓ Raw data table
- ✓ Site map/photo permission
- ✓ Calculation units
- ✓ Source references

## Analysis

- ✓ Source–pathway–receptor
- ✓ Cause analysis
- ✓ At least two alternatives
- ✓ Safety and legal awareness
- ✓ Limitations

## Recommendation

- ✓ Responsible person
- ✓ Timeline
- ✓ Resource need
- ✓ Monitoring indicators
- ✓ Expected benefit

# Practice Questions and Environmental Problem Laboratory

Searchable short answers, diagrams, engineering scenarios and numerical practice.

## Module 1 practice

1. Draw an ecosystem showing energy and nutrient flow.
2. Compare pyramid of number, biomass and energy.
3. Create a food web with at least eight organisms.
4. Explain lentic and lotic systems using local examples.
5. Draw carbon and nitrogen cycles.
6. Explain how fertiliser changes nitrogen/phosphorus cycles.
7. Differentiate greenhouse effect and global warming.
8. Explain ozone depletion mechanism and effects.
9. Trace acid-rain formation from fuel combustion.
10. Calculate energy at four trophic levels using an assumed transfer percentage.

## Module 2 practice

1. Apply source–pathway–receptor to a roadside workshop.
2. Classify ten pollutants as particulate/gaseous and primary/secondary.
3. Draw cyclone, bag filter and ESP.
4. Select control equipment for cement dust, boiler dust and soluble gas.
5. Calculate outlet concentration for 95% removal.

6. Compare pH, DO, BOD and COD.

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7. Interpret a river sample with DO 3 mg/L and BOD 20 mg/L.

8. Calculate BOD reduction from 250 to 25 mg/L.

9. Prepare source-path-receiver controls for noise.

10. Combine two equal 90 dB sources.

## Module 3 practice

1. Label a flat-plate collector.

2. Compare liquid and air collectors.

3. Explain solar pond, dryer and still.

4. Calculate heat for 200 L water through 40°C rise.

5. Explain biomass fuel properties.

6. Draw anaerobic-digestion stages.

7. Estimate biogas from a given daily feed.

8. Calculate wind power using given data.

9. Compare wind benefits and problems.

10. Compare hydrogen, wave, tidal, OTEC and geothermal systems.

## Module 4 practice

1. Characterise municipal, electronic and biomedical waste.

2. Prepare a segregation plan for a campus.

3. Develop a material balance for a workshop.

4. Explain 3R with engineering examples.

5. Draw a sanitary-landfill cross-section.

6. Calculate landfill volume from mass and compacted density.

7. Prepare a simple carbon-footprint inventory.

8. Create an aspect–impact register for welding and painting.

9. Explain PDCA for ISO 14001.

10. Prepare a complete case-study proposal.

## Challenge problems with hints

### 1. Ecological energy

A producer level has 50,000 kJ and transfer is 12% per trophic level. Find energy at three consumer levels. **Hint:** multiply by 0.12 repeatedly.

### 2. Air-control efficiency

Inlet dust is 800 mg/m<sup>3</sup> and outlet 32 mg/m<sup>3</sup>. Find efficiency. **Hint:** (inlet – outlet)/inlet × 100.

### 3. Water treatment

BOD falls from 320 to 48 mg/L. Find removal and discuss whether one value proves compliance. **Hint:** calculate 85%; compliance also needs current standard and method.

### 4. Solar heat

Heat 250 kg water from 28°C to 62°C. **Hint:**  $Q = mc\Delta T$ .

### 5. Wind speed change

All else equal, compare power at 6 and 9 m/s. **Hint:** ratio =  $(9/6)^3 = 3.375$ .

### 6. Waste diversion

Waste = 500 kg/day; 45% composted and 30% recycled. Find disposal fraction. **Hint:** 25% or 125 kg/day.

### 7. Carbon reduction

Electricity decreases from 1,200 to 900 kWh/month at 0.65 kg CO<sub>2</sub>e/kWh. **Hint:**  $300 \times 0.65$ .

## Common mistakes and corrections

| Mistake                                            | Correction                                                               |
|----------------------------------------------------|--------------------------------------------------------------------------|
| Calling every environmental change “pollution”     | Identify adverse change, pollutant, concentration/duration and receptor. |
| Arrow reversed in food chain                       | Arrow follows energy transfer from food to consumer.                     |
| Saying energy cycles                               | Energy flows and dissipates; matter cycles.                              |
| Confusing ozone depletion with global warming      | Explain separate gases, mechanisms and effects.                          |
| Assuming COD and BOD are identical                 | State the different test principles and what each indicates.             |
| Adding decibels arithmetically                     | Use logarithmic addition.                                                |
| Quoting renewable energy as impact-free            | Discuss life-cycle, siting, material and operational impacts.            |
| Treating mixed waste as recyclable                 | Segregate clean streams at source.                                       |
| Using outdated law names without verification      | Check current official text and amendments.                              |
| Giving a case-study solution without baseline data | Measure/observe first, then compare alternatives.                        |

## Quick Revision Map

Compact memory cues for all four modules.

### Ecosystem

Abiotic + biotic interactions.

### Energy

Flows; nutrients cycle.

### Food web

Interconnected food chains.

**Lentic**

Standing water.

**Lotic**

Flowing water.

**Carbon cycle**

Photosynthesis, respiration, decomposition, combustion.

**Nitrogen**

Fixation → nitrification → assimilation → ammonification → denitrification.

**Global warming**

Enhanced greenhouse effect.

**Acid rain**

Mainly SO<sub>2</sub>/NO<sub>x</sub> transformation.

**Pollution model**

Source → pathway → receptor.

**Cyclone**

Centrifugal separation.

**Bag filter**

Fabric filtration.

**ESP**

Electrical charging/collection.

**DO**

Dissolved oxygen.

**BOD**

Biological oxygen demand.

**COD**

Chemical oxygen demand.

**Noise**

Logarithmic dB scale.



### **Solar collector**

Cover + absorber + tubes + insulation.

### **Biogas**

Four anaerobic stages.

### **Wind**

$P \propto v^3$ .

### **Hydrogen**

Energy carrier.

### **3R**

Reduce, reuse, recycle.

### **Landfill**

Liner, leachate, gas, cover, monitoring.

### **Carbon footprint**

Activity  $\times$  factor.

### **EMS**

Plan–Do–Check–Act.

### **Case study**

Baseline  $\rightarrow$  alternatives  $\rightarrow$  recommendation  $\rightarrow$  indicators.

**Revision order:** Ecosystem  $\rightarrow$  cycles  $\rightarrow$  atmospheric problems  $\rightarrow$  pollution devices  $\rightarrow$  water/noise parameters  $\rightarrow$  renewable systems  $\rightarrow$  waste hierarchy  $\rightarrow$  carbon/EMS  $\rightarrow$  case-study method.

## **Forty Important Questions and Answers**

### **1. What is an ecosystem?**

A community of organisms interacting with the physical and chemical environment.

### **2. Producer?**

An organism that makes organic matter from inorganic materials using an energy source.

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### 3. Decomposer?

An organism that breaks down dead matter and returns nutrients.

### 4. Food web?

A network of interconnected food chains.

### 5. Why is energy pyramid upright?

Energy is lost as heat at every trophic transfer.

### 6. Lentic versus lotic?

Lentic is standing water; lotic is flowing water.

### 7. Nitrogen fixation?

Conversion of atmospheric nitrogen into biologically usable compounds.

### 8. Eutrophication?

Nutrient enrichment causing excessive growth and possible oxygen depletion.

### 9. Greenhouse effect?

Atmospheric absorption and re-emission of outgoing infrared energy.

### 10. Acid rain precursor?

Mainly sulphur dioxide and nitrogen oxides.

### 11. Pollutant?

A substance or energy form capable of adverse effect at a given concentration/duration.

### 12. Primary pollutant?

Emitted directly from a source.

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### 13. Cyclone principle?

Centrifugal separation of particles from gas.

### 14. Bag-filter principle?

Particles are captured as gas passes through fabric.

### 15. ESP principle?

Charged particles migrate to collection plates.

### 16. Absorber?

A device transferring a gas pollutant into a liquid.

### 17. Catalytic converter?

A vehicle exhaust device catalysing reactions that reduce CO, hydrocarbons and NO<sub>x</sub>.

### 18. DO?

Dissolved oxygen available in water.

### 19. BOD?

Oxygen used biologically to degrade biodegradable matter under test conditions.

### 20. COD?

Oxygen-equivalent measure of chemically oxidisable material under the test.

### 21. Why not add dB directly?

The decibel scale is logarithmic.

### 22. Flat-plate absorber?

A dark surface converting solar radiation to heat.

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### **23. Selective coating?**

High solar absorption with relatively low thermal emission.

### **24. Four digestion stages?**

Hydrolysis, acidogenesis, acetogenesis and methanogenesis.

### **25. Main biogas component?**

Methane, with carbon dioxide and minor constituents.

### **26. Wind power relation?**

Available power is proportional to air density, swept area and wind speed cubed.

### **27. Hydrogen source or carrier?**

Generally an energy carrier.

### **28. Tidal energy advantage?**

Tides are highly predictable.

### **29. Geothermal binary cycle?**

Geothermal fluid heats a separate low-boiling working fluid.

### **30. E-waste?**

Discarded electrical/electronic equipment and components.

### **31. 3R?**

Reduce, reuse and recycle.

### **32. Leachate?**

### 33. Hazardous waste?

Waste with dangerous properties such as toxicity, corrosivity, reactivity or flammability.

### 34. Carbon footprint?

Total greenhouse-gas emissions within a defined boundary, expressed as CO<sub>2</sub>e.

### 35. Carbon credit?

A quantified reduction/removal unit under a defined programme.

### 36. Environmental aspect?

An element of an activity, product or service that interacts with the environment.

### 37. Environmental impact?

A change to the environment resulting from an aspect.

### 38. ISO 14001?

A standard specifying requirements for an environmental management system.

### 39. PDCA?

Plan, Do, Check and Act.

### 40. First case-study step?

Define an approved problem, objective, boundary and safe data method.

# Series-Test Answers, Solved Challenge Problems and References

## Series Test I answer guidance

1. **Ecosystem:** define community–environment interaction; list biotic producers/consumers/decomposers and abiotic water, air, soil, light and temperature.
2. **Food chain/web/pyramid:** give definitions, arrow direction, one example and explain energy loss.
3. **Cycle:** draw reservoirs and labelled processes; carbon or nitrogen earns strong coverage.
4. **Atmospheric problems:** explain different mechanism, gases, effects and controls.
5. **Pollution:** define pollutant and show source–pathway–receptor.
6. **Control device:** include construction, working principle, application, advantage and limitation.
7. **DO/BOD/COD:** distinguish oxygen present, biological demand and chemical demand.
8. **BOD removal:**  $(240 - 36)/240 \times 100 = 85\%$ .
9. **Noise:** sources, effects, source/path/receiver controls and regulatory awareness.
10. **Two 85 dB sources:** approximately 88 dB.

## Series Test II answer guidance

1. **Flat plate:** transparent cover, absorber/coating, flow passage, insulation, casing and heat-transfer process.
2. **Digestion:** explain four stages, gas composition, uses, digestate and safety.
3. **Wind numerical:** use  $\frac{1}{2}\rho A v^3 C_p$  with units and state simplifications.
4. **New sources:** compare origin, conversion, application and challenge.
5. **E/biomedical waste:** sources, characteristics, risk, segregation and authorised management.
6. **3R/landfill:** hierarchy plus liner, leachate, gas, cover and monitoring.
7. **Carbon:** footprint is inventory; credit is verified reduction/removal unit.
8. **Fabrication:** include activity, aspect, impact, control and indicator.
9. **ISO:** Plan–Do–Check–Act with aspects, objectives, controls, audit and improvement.
10. **Case study:** problem, baseline, method, alternatives, recommendation and indicators.

## Challenge-problem answers

| Problem                | Answer                                                                                             |
|------------------------|----------------------------------------------------------------------------------------------------|
| Ecological energy      | $50,000 \times .12 = 6,000 \text{ kJ}; \times .12 = 720 \text{ kJ}; \times .12 = 86.4 \text{ kJ}.$ |
| Air-control efficiency | $(800 - 32)/800 \times 100 = 96\%.$                                                                |
| BOD removal            | $(320 - 48)/320 \times 100 = 85\%;$ compliance requires current standard and method.               |
| Solar heat             | $Q = 250 \times 4.18 \times 34 = 35,530 \text{ kJ} = 35.53 \text{ MJ}.$                            |
| Wind ratio             | $(9/6)^3 = 3.375;$ theoretical available power is 3.375 times.                                     |
| Waste diversion        | 75% diverted; 25% disposed = 125 kg/day.                                                           |
| Carbon reduction       | $(1,200 - 900) \times 0.65 = 195 \text{ kg CO}_2\text{e/month}.$                                   |

# Text and reference list from the syllabus

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1. C. N. R. Rao — *Understanding Chemistry*, Universities Press, 2011.
2. William Nazaroff and Lisa Cohen — *Environmental Engineering Science*.
3. C. S. Rao — *Environmental Pollution Control and Engineering*.
4. M. N. Rao and H. V. N. Rao — *Air Pollution*.
5. Frank Kreith and Jan F. Kreider — *Principles of Solar Engineering*.
6. Aldo Vieira da Rosa — *Fundamentals of Renewable Energy Processes*.
7. A. D. Patwardhan — *Industrial Solid Waste*.
8. Metcalf & Eddy — *Wastewater Engineering*.

The syllabus also lists environmental portals including TERI, pollution-control resources, India Environment Portal, United Nations Sustainable Development Goals and general energy-conservation resources. Verify current official web addresses before use because legacy syllabus URLs may contain typographical or domain changes.

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Author: Nandakumar.M • [polypmna.dpdns.org](http://polypmna.dpdns.org)